SmartLink Software Design

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1. AUTHENTICATION REQUIREMENTS

ID	Functional Requirement	Description	Input	Output	Priority

1.1	**User Sign Up (Basic Credentials)**	**The system should allow users to register with the following information:** **Required Fields** **• Full Name** **• Email (unique)** **• Password** **• Birth Date** **• Phone Number (unique)** **• Gender** **Validation Rules** **Email Validation** **Format: [username]@[domain]** **Length: 6–40 characters** **Pattern: Must follow a valid email structure** **Uniqueness: Email must not already exist in the system** **Examples:** **✅ Valid: user@example.com, john.doe@company.co.uk** **❌ Invalid: user@, @domain.com, user domain.com** **Password Validation** **Length: 8–20 characters** **Required Character Types:** **- English letters (A–Z, a–z)** **- Numbers (0–9)** **- Symbols: ! @ # $ % ^ & * ( ) _ + - { } [ ] \ : ; " ' < > , . ? /** **Strength Requirements:** **- At least one uppercase letter** **- At least one lowercase letter** **- At least one number** **- At least one special character** **Examples:**	Full name Email (unique) Password Birth date (required) Gender (required)	**Success:** - Send verification email with navigation link - Notify user to check email **Failure:** - Display error message - Prompt to retry with valid information	**HIGH**

		✅ Valid: Pass123!, MyP@ssw0rd, Secure#456 ❌ Invalid: password (no uppercase/number/symbol), Pass123 (no symbol) Confirm Password Validation Rule: Must exactly match the password field Error Message: "Passwords do not match" Full Name Validation Length: 5–50 characters Allowed Characters: Letters (A–Z, a–z), spaces, hyphens (-), apostrophes (') Trimming: Leading and trailing spaces removed Examples: ✅ Valid: John, Mary-Jane, O'Connor ❌ Invalid: J, John123, @John Birth Date Validation Format: YYYY-MM-DD Age Requirements: - Minimum: 13 years - Maximum: 120 years Date Range: Must be a past date (no future dates) Examples: ✅ Valid: 1990-05-15 (age 35) ❌ Invalid: 2030-01-01 (future date) Gender Validation Type: Dropdown or Radio Button Options: - Male			

		- Female **- Prefer not to say** **Required: User must select one option** **Error Handling** **Invalid fields display red error messages around the input area.** **Form Buttons** **• Sign Up** **• Reset** **Additional Option** **Already have an account? → Login**			
1.2	**User Sign Up (Gmail)**	Users can sign up using Gmail OAuth integration to create a new account. Same sign-up form as 1.1 with Google authentication service integration.	User Gmail credentials	**Success:** - Display success notification - Navigate to home page **Failure:** - Display error message (e.g., user already exists) - Prompt to retry	**HIGH**

1.3	**User Login**	Users can log in to their account using basic credentials (email and password) or Gmail. **Form Fields:** - User email - Password **Validation:** Same validation rules as requirement 1.1 **Error Handling:** Red error messages for invalid fields. Data cannot be submitted if validation fails. **Form Buttons:** - Sign in - Sign-in using Gmail - Don't have an account? Sign Up	(Email and password) OR User Gmail credentials	**Success:** - Display success notification - Navigate to home page **Failure:** - Display error message (user not found/incorrect password) - Prompt to retry with valid credentials	**HIGH**

1.4	**Admin Sign In**	The system provides a secure, separate authentication portal for administrators. Admin users have elevated privileges to manage users, moderate content, and access analytics. Admin Features: - Hardcoded super-admin account for initial setup - Ability to promote other users to admin role - Dedicated admin login form (separate from user login) Admin Capabilities: - User account management - Content moderation - Platform analytics access - System configuration	Admin username and password	**Success:** - Navigate to admin dashboard **Failure:** - Display error message for invalid credentials	**HIGH**

2. JOB SEEKERS / PROFESSIONALS

ID	Functional Requirement	Description	Input	Output	Priority

2.1	**Profile Management**	**User Profile Management** The system should allow users to view and manage their profile information through a dedicated profile page. **Profile Page Overview** • The profile page appears when the user clicks the Profile Icon in the navigation bar or user menu. • Users can view or update their information using the Edit Buttons provided beside each field. **Editable Profile Information** Users can view and edit the following details: 1. First and Last Name 2. Professional Headline (e.g., “Software Engineer”, “UI/UX Designer”) 3. Bio / Summary 4. Skills (add, remove, or modify) 5. Experience (job titles, companies, durations) 6. Education (institutions, degrees, years attended) **Field Constraints and Validation Rules** **Name Fields** • Length:2 –50 characters • Allowed Characters: Letters (A–Z, a–z) and spaces only • Trimming: Leading and trailing spaces are removed Examples: ✅ Valid: John Smith, Mary Anne ❌ Invalid: John123 **Professional Headline**	Profile data fields	- Updated profile information - Success/error message after save - Real-time validation feedback	**HIGH**

		• Length: 3–100 characters • Allowed Characters: Letters, numbers, spaces, commas, hyphens, and ampersands (&) Examples: ✅ Valid: Front-End Developer, Marketing & Communications Lead ❌ Invalid: $$ Coder $$ **Bio / Summary** • Length: Up to 500 characters • Allowed Characters: All standard printable characters • Purpose: Allows users to briefly describe themselves, career goals, or interests **Skills** • Input Method: Tag-based system with auto-suggestions (e.g., “JavaScript”, “Python”, “Project Management”) • Add/Remove Options: Users can add multiple skill tags or delete existing ones • Character Limit per Skill: Up to 30 characters Examples: ✅ Valid: C++, Data Analysis, Web Development **Experience** • Fields: Job Title, Company Name, Duration, Description (optional) • Constraints: - Job Title and Company Name: 1–100 characters - Duration: Must include valid start and end dates (YYYY-MM format) - Description: Up to 300 characters **Education** • Fields: Institution Name, Degree, Field of Study, Years Attended			

		• Constraints: - Institution Name and Degree: 1–100 characters - Years Attended: Must be in the format YYYY–YYYY and within logical date ranges **Error Handling** • Invalid inputs display red error messages near the affected field. • Changes are saved only after all validations pass successfully. **Available Actions** • Edit Profile: Enables editing mode for all fields. • Save Changes: Updates the user's profile after successful validation. • Cancel: Discards unsaved edits and restores the previous profile view.			

2.2	**Professional Feed**	**Feed Overview:** A dynamic, personalized feed showing posts from the user's professional network in reverse chronological order. **Feed Features:** ● Posts from direct connections ● Sorted by newest first ● Updates on refresh **Post Card Includes:** ● Profile picture and name (clickable) ● Text or media content ● Timestamp (e.g., "2h ago") ● Like and comment buttons with counts **Loading:** ● Loads first 20 posts initially for example ● Infinite scroll or "Load More" for next 20 ● Loading indicator during fetch **Engagement:** ● Like/unlike directly ● Expand comments section	None	List of posts from connections with user interactions	**HIGH**

2.3	**Post Creation**	Users can create and publish professional posts to share updates, ideas, or achievements within their network. Post Creation Form The form provides a text area supporting up to 1000 characters for post content, along with an option to upload images to make posts more engaging. Once the content is ready, users can publish it using the “Post” button. Post Features The system validates all inputs to ensure posts are not empty and that uploaded media follow the required format and size limits, maintaining content quality and consistency across the platform.	Post content and optional media	- Post published to feed - Success confirmation - Post appears in user's feed and connections' feeds	**HIGH**
2.4	**Post Interactions**	Users can like and comment on posts. **Each Post Features:** - Like button (heart icon) with toggle functionality - Comment button that expands comment section - Comment input field with submit button	Like action OR comment text	- Updated like count - Added comment displayed - Success feedback	**HIGH**

2.5	**POST MODIFICATION**	Users should be able to modify the content of a post . if the post is a user's post ,the user will see a modification button to change the content.	-click the edit button -a compose page similar to create will appear with the content of the post. -click confirm to edit the post	-the post will be edited in the database -the post will be edited to all other users	**MEDIUM**

2.6	**Connection Management**	Users can manage professional connections. **Connection Management Page:** - List of connection requests received - List of connection requests sent - List of current connections **For Each Request:** - Accept button - Decline button **For Each Connection:** - Remove connection option - View profile link Search functionality to find users to connect with.	Connection actions (send/accept/decline/remove)	- Updated connection status - Success/error messages	**HIGH**

2.7	**Search Functionality**	Users can search for other professionals and jobs. **Search Page Features:** - Search bar with auto-suggestions - Filters: - Users/Jobs toggle - Skills filter (multi-select) - Location filter - Experience level - Company size (for jobs) - Industry **User Search Results:** - Name, headline, current position - Common connections - Connect button **Job Search Results:** - Job title, company, location - Required skills - Apply/Save buttons	Search keywords and filter criteria	- Filtered list of users/jobs - Number of results found - Relevant matching information - Pagination for large result sets	**HIGH**

3. RECRUITERS / EMPLOYERS

ID	Functional Requirement	Description	Input	Output	Priority
3.1	**Job Posting Management**	Recruiters can create and manage job postings. **Job Posting Form Fields:** - Job title (required) - Company name (required) - Job description (min 100 chars, max 2000 chars) - Required skills (tag-based input) - Nice-to-have skills - Experience level (Entry/Mid/Senior) - Job type (Full-time/Part-time/Contract/Internship) - Location (On-site/Remote/Hybrid) - Salary range (optional) - Application deadline **Features:** - Form validation with real-time feedback - Draft saving capability	Job details and requirements	- Job posted successfully message - Job appears in job listings - Application link generated - Edit/delete options for posted jobs	**MEDIUM**

3.2	**Candidate Search**	Recruiters can search and filter candidates. **Search Interface:** - Search by skills (multiple selection) - Filter by: (Expected example) - Experience level - Education level - Location preference - Availability - Salary expectations **Search Results Display:** - Candidate name and headline - Skills match percentage - Current position - Connect/Message buttons - Quick view profile option	Search criteria and filters	- List of matching candidates - Candidate contact information - Export results option (CSV)	**MEDIUM**

4. ADMINISTRATORS

ID	Functional Requirement	Description	Input	Output	Priority

4.1	**User Account Management**	Admin can manage user accounts and roles. **Admin Dashboard Features:** - List of all users with pagination - Search users by name/email - Filter by user type (Job Seeker/Recruiter/Admin) **Actions for Each User:** - Deactivate/Activate account - View full profile	User selection and action type	- Updated user roles/status - Success/error notifications - Audit log of admin actions	**HIGH**

4.2	**Content Moderation**	Admin can manage content and moderate posts. **Content Management Dashboard:** - Recently reported posts - All posts with filtering options **Available Actions:** - View post details - Remove inappropriate posts - Warn post author - Ban repeat offenders **Features:** - Moderation queue for reported content - Quick action buttons for common tasks	Content selection and moderation action	- Content removed/approved - User notifications sent - Success confirmation	**MEDIUM**

5. SYSTEM FEATURES

ID	Functional Requirement	Description	Input	Output	Priority

5.1	**Personalized Feed Generation**	System generates personalized feed for users. **Feed Generation Algorithm Based On:** - User's connections - User's skills and interests - Recency of posts **Feed Includes:** - Posts from direct connections - Popular posts from extended network - Relevant job postings - Industry news suggestions	User profile and engagement data	- Personalized post feed - Relevant content suggestions - Engagement metrics - Fresh content indicators	**HIGH**

5.2	**Job Recommendations**	Users can receive personalized job recommendations using AI matching. **Matching Based On:** - User profile skills - User experience - Education background - Previous job applications **Job Recommendation Card:** - Job title and company - Job location (remote/onsite/hybrid) - Salary range (if available) - Apply button **User Actions:** - Click "Not interested" to improve recommendations - Filter by location, job type, experience level	User profile data and preferences	- List of personalized job recommendations - Match percentage for each job - Option to apply or save jobs	**MEDIUM**

5.3	**Resume Analysis**	Users can upload resumes and receive AI-generated improvement tips. **Resume Analyzer Features:** - File upload area (PDF, DOC,.....) **Analysis Provides:** - Overall resume score (0-100) - Grammar and spelling feedback - Formatting suggestions - Content completeness assessment - Skill highlighting recommendations	Resume file (PDF/DOC)	- Resume analysis report - Improvement suggestions - Overall quality score - Downloadable report	**MEDIUM**

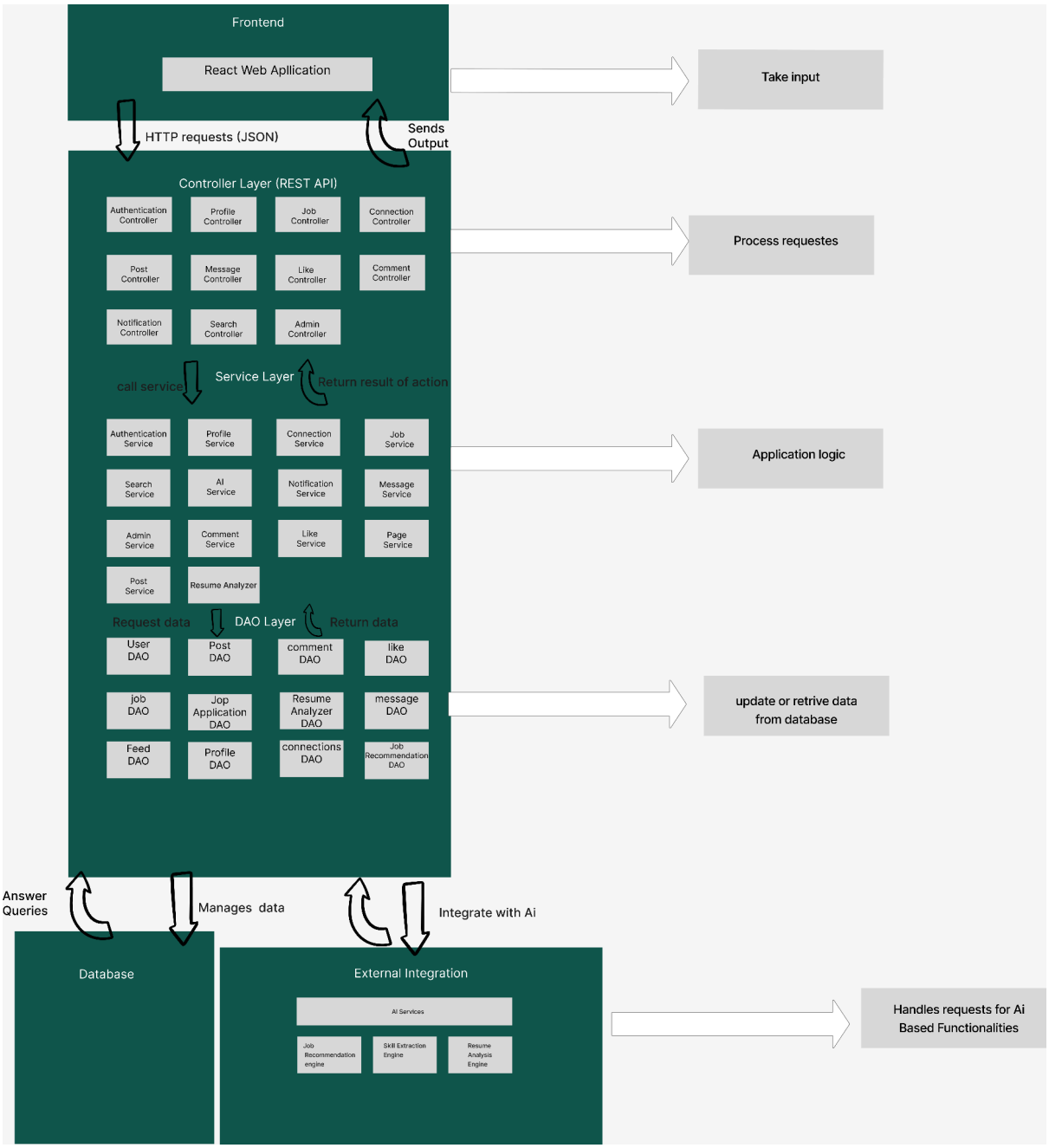
5.4	**Skill Extraction**	System automatically extracts skills from uploaded resumes. **Extraction Form Features:** Integrated with resume upload. System automatically detects: - Technical skills - Soft skills - Programming languages - Tools and technologies - Certifications **Extracted Skills Display:** Skills shown as tags that user can: - Confirm to add to profile - Remove if incorrect - Add missing skills manually Confidence score shown for each extracted skill.	Resume file	- List of extracted skills with confidence scores - Option to add confirmed skills to profile - Success message after profile update	**MEDIUM**

NON-FUNCTIONAL REQUIREMENTS

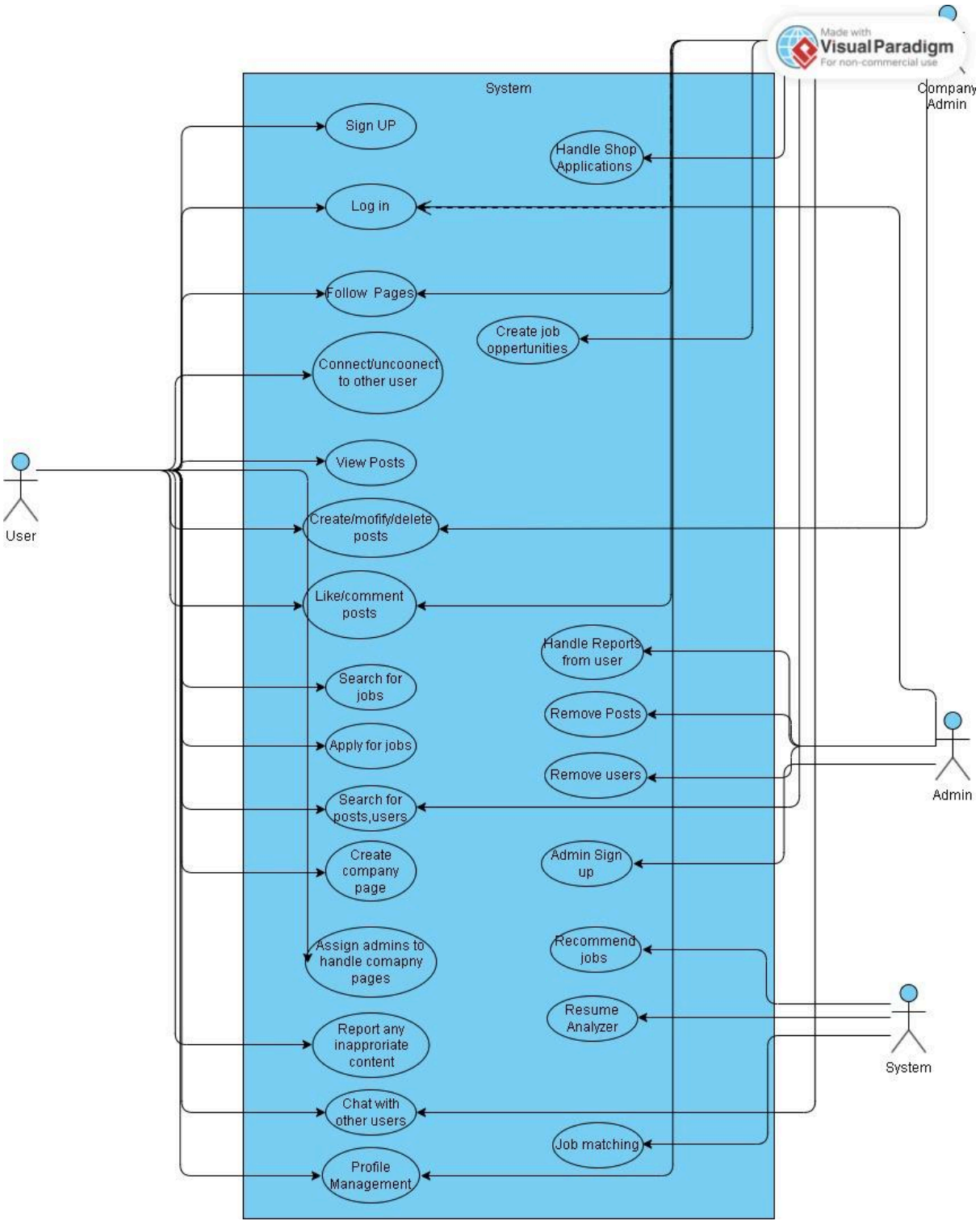
ID	Category	Description	Priority
NF1	**Security**	- Passwords must be **encrypted** and users must authenticate before accessing private pages. - Secure file upload validation	**HIGH**
NF2	**Usability**	The user interface should be **intuitive and easy to navigate**, with clear labels and simple forms focus on accessibility and simplicity.	**HIGH**
NF3	**Scalability**	- The system design allow adding more modules (e.g., Messaging, Notifications) later without major changes Focus on flexible architecture.	**MEDIUM**
NF4	**Maintainability**	- Modular code structure - Comprehensive documentation - Easier to update and fix in the future.	**MEDIUM**
NF5	**Compatibility**	- Cross-browser support (Chrome, Firefox, Safari, Edge)	**LOW**

2. System Architecture & Main Components

[Components – Figma]



3. **Use Case Diagrams**



4. **Interaction Diagrams for Main Functionalities**

User Sequence Diagram.pdf

Admin Sequence Diagram.pdf

Company Sequence Diagram.pdf

5. **Detailed Class Diagram**

SmartLink – FigJam

6. **Relational Data Model**

RCU_Schema (Community) – FigJam

7. **CRC Cards (Class Responsibility Collaborator)**

CRC Card Template (Community) – FigJam

8. Design Patterns

1) **Singelton (one Email for account)**
2) **Builder (for Profile)**
3) **State (for user and admins)**
4) **Observer (for notification)**
5) **Strategy (for different options of sign in)**
6) **Filter (for search)**
7) **Iterator (for connections)s**



[Untitled Workspace - Creately]

8. Changes to Project Proposal

NOTIFICATIONS SYSTEM (Optional)

ID	Functional Requirement	Description	Input	Output	Priority
6.1	User Notifications	Simple notification system that alerts users about key activities. Triggers: - New connection requests - Accepted connections - New messages (if messaging enabled) - Job application updates Interface: - Notification icon displaying count badge - Dropdown list of recent notifications	User activities	- Notification count badge - List of notifications - Read status updates	MID

6.2	**Email Notifications**	**Basic email alerts triggered by important system events.** **Features:** **- Welcome email after signup** **- New connection request email** **- Security alert notifications** **Email Templates:** **Plain text or simple HTML emails with minimal branding and clear CTAs.**	**User events**	**- Email sent confirmation** **- Delivery tracking (basic)**	**LOW *(Optional)***

MESSAGING SYSTEM (Optional)

ID	Functional Requirement	Description	Input	Output	Priority

7.1	**Basic Direct Messaging**	**Simple internal messaging feature for connected users.** **Capabilities:** **- Send and receive text messages** **- Conversation history stored per user pair**	**Message text**	**- Message sent confirmation** **- Messages visible in conversation history** **- Error message on failed delivery**	**MID**

7.2	**Message Management**	**management tools for private conversations.** **Functions:** **- Delete individual or all conversations** **- Block users from messaging** **Interface:** **Simple buttons for control actions within message list**	**User actions**	**- Conversation removed** **- User block confirmation** **- Report successfully sent**	**LOW *(Optional)***

MILESTONE SCHEDULE

Milestone 1: Project Setup & Core Authentication

Objectives: Establish the foundation of the system by setting up the core environment, database, and authentication modules.

Deliverables:

- Development environment setup (React frontend + Spring Boot backend)
- Database schema design and connection setup
- **User Authentication Module:**
    - User Sign Up (Basic Credentials & Gmail) *(Req. 1.1, 1.2)*

    - User Login (Email/Gmail) *(Req. 1.3)*
    - Admin Sign In *(Req. 1.4)*

- **Profile Management (Basic view/edit)** *(Req. 2.1)*
- **Professional Feed System** *(Req. 2.2, 5.1)*
- **Post Management**
    - Post Creation *(Req. 2.3)*
    - Post Interaction (Like & Comment) *(Req. 2.4)*
    - Post Modification *(Req. 2.5)*

Optional (if time allows):

- *Basic Email Notification after signup (Req. 6.2)*

Milestone 2: Core Networking Features

Objectives: Build the platform's interactive and social components, enabling users to connect, share, and interact.

Deliverables:

- **Connection Management System** *(Req. 2.6)*
- **Search Functionality (Users/Jobs)** *(Req. 2.7)*
- **Recruiter Features**
    - Job Posting Management *(Req. 3.1)*
    - Candidate Search *(Req. 3.2)*
- **Content Moderation (Admin Dashboard)** *(Req. 4.2)*

Optional (if time allows):

- In-app Notification System *(Req. 6.1)*

- Email Notifications for job updates or security alerts *(Req. 6.2)*

Milestone 3: AI Features & Advanced Functionality

Objectives: Integrate intelligent and data-driven modules that enhance personalization, recruitment, and system scalability.

Deliverables:

- **AI-Based Job Recommendations** *(Req. 5.2)*
- **Resume Analysis System** *(Req. 5.3)*
- **Skill Extraction Automation** *(Req. 5.4)*
- **System Analytics and Admin Enhancements** *(Req. 4.3)*

Optional (if time allows):

- **Basic Direct Messaging**
- Message Management (delete/block/report) *(Req. 7.2)*